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Question: 1

Define array:-

An array is collection of similar data types stored in contiguous location and accessed using a index.

Example:-

int Marks[5] = {10, 20, 30, 40, 60};

2 Difference from simple val

Feature	simple variable	Array
Storage	stores one vale	stores Multiple vale
Memory	sigle Memory block	contiguous mutiple blocks
access	Direct	indexed

Question No 2 :-

Difference between 1D and 2D array

Feature	1D Array	2D array
Defination	Linear list of elements	Table rowxcolus list of elements

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Syntax	Int arr[5]	int arr[3][3]
Access	arr[2]	arr[1][1]
visualization	1, 2, 3, 4	1 2 3 4 5 6

!
1D → Linear storage
2D → Grid like storage.

Question 3:- how memory allocation

Arrays occupy continuous memory blocks so the CPU can access elements using formula.

Formula:-

arr[i] = Base address + (i * size of each element)

Example:-

array[5] = {10, 20, 30, 3.}

This makes fast access in O(1)

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Question 4:- syntax of arrays

1D Array syntax:-

```
int marks[5]
int marks[5]={10,20,30,40}
```

2D array syntax:-

```
int marks[3][3];
int marks[3][3]={{1,2,3},
                 {4,5,6},
                 {7,8,9}}
```

Question 5 :- what is Pointer?

A Pointer is variable that stores memory address of another elements

```
int X=10
int *ptr=&X
```

Accessing array element with Pointers

```
int arr[3]={10,20,30}
int *P=arr;
cout<<*(P+0)
cout<<*(P+1); 20
```

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Question 6:-

In C++, array name behave like a pointer to its first element

```
int arr[4]={5,4,10,50};
cout<<arr;
cout<<*arr;
cout<<*(arr+2);
```

Key relationship:-

arr[i] = *(arr+i)

Explanation:-

- arr gives the address of first element.
- arr+i moves the pointer i elements ahead.
- *(arr+i) access the value

Question 7:-

- Dynamic memory allocation is the process of allocating memory at runtime instead of compile time. It allows flexible memory usage base on program needs.

List:-

Malloc(), realoc() calloc()
free()

Name : Ghulam Habib
Roll No : 24-NTU-CS-FL-1154

Question 8

Dynamic memory allocation related functions:-

malloc() → allocate signle block of memory.

calloc() → Allocates multiple and initize the to zero

Realloc() → Resize Previarly allocated me or

free() free the allocated memory

```
int *ptr =(int *) malloc(5* sizeof(int)
```

Question 9:-

Insertion mean addings a new element at a spesific Position.

Insertion at beginig:-

- shift all elements to right Post
- insert new value

```
for(int i=n; i>0; i--
    arr[i] = arr[i-1]
arr[0] = value
```

Insertion between:-

shift element to right by one position

Name: Ghulam Habib
Roll No: 24-NTU-CS-FL-1184

```
for (int i=n; i>pos ; i--)
        arr[i] = arr[i-1]
arr[pos] = newvalue
```

insertion at end.

```
simple arr(n)
arr[n] = vale
n++
```

Question 10:-

Deletion means removing an element and shifting remaining ones to fill the gap.

a **Deletion from beginning:-**

Moves all element to one left:

```
for(int i=0; i<n-1; i++)
        arr[i] = arr[i+1]
n--;
```

b) **Deletion from in between**

Shift all elements after that position to left.

```
for (int i = pos; i<n-1; i++)
        arr[i] = arr[i+1]
n--;
```

Name: Ghulam Habib
RollNo: 24-NTU-CS-FL-1154

Deletion from end:-

just reducing the array size by:

```
n--;
```